

Beyond the Hype

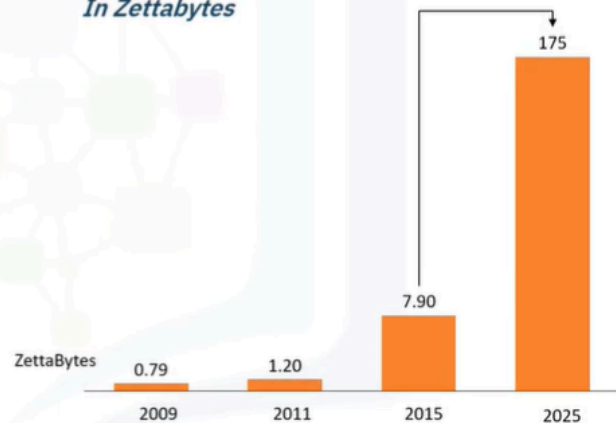
The hype about Big Data

Facts:

- More data has been created in the past two years than in the entire previous history of humankind
- 40% projected growth in global data generated per year
- The amount of digital data created over the next five years will be greater than twice the amount of data created since the advent of digital storage

Sources: McKinsey report – Digital era, data growth, estimates and, Gartner, IDC estimates

Growth in global data *In Zettabytes*



- So why is everyone talking about Big Data? More data has been created in past two years than in the entire previous history of humankind. There is a 40% projected growth in global data generated per year. By 2026, the amount of digital data created will exceed twice the amount of data created since the advent of digital storage. The chart on the right shows the growth in global data in zettabytes.

Where does Big Data come from?

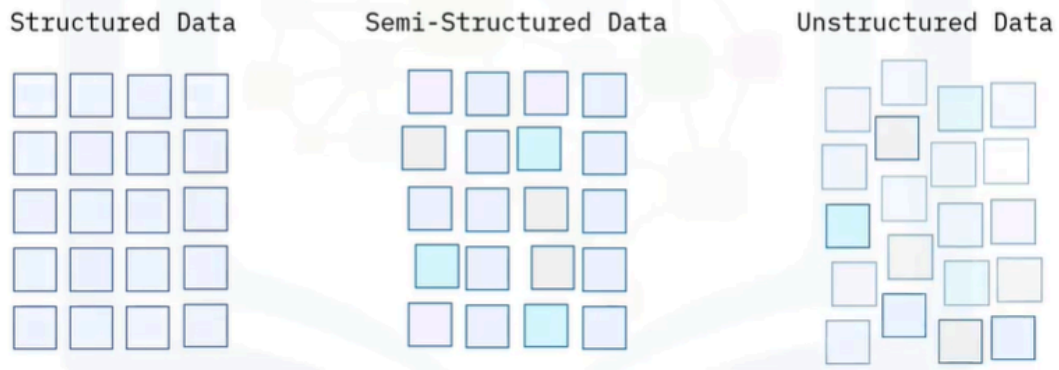
- Social data
- Machine data
- Transactional data



- There are three major sources of Big Data. Social data comes from the likes, Tweets & Retweets, comments, video uploads, and general media that are uploaded and shared via the world's favorite social media platforms. Machine-generated data and business-generated data, are data that organizations generate within their own operations. Machine data is defined as information which is generated by Internet of Things (IoT) sensors equipped industrial equipment that are installed in machinery, and even web logs that track user behavior. Transactional data is generated from all the daily transactions that take place both online and offline, such as invoices, payment orders, storage records, delivery receipts.

Types of Big Data

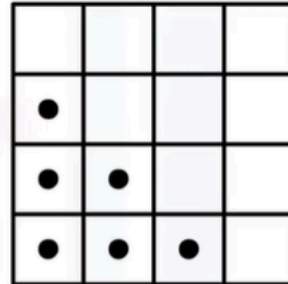
- Structured, unstructured, semi-structured



- Big Data comes in three types: structured, semi-structured, and unstructured data. Structured data is data that is organized, labeled, and has a pre-defined data model that conforms to a tabular format.
- Common examples of structured data include names, dates, addresses, and credit card numbers. Unstructured data is said to make up about 80% of data in the world and does not conform to any predetermined format. It includes images, audio or video files, sensor data from IoT devices, text messages, and social media posts.
- Semi-structured data includes components of both structured and unstructured data in that it is mostly unstructured, but includes some metadata that identifies certain characteristics. Emails fall into the category of semi-structured data because, while the content in an email is usually unstructured, there are several characteristics such as the email addresses, subject line, logo images, and URLs that can be pulled into structured tables.

Sources: Structured

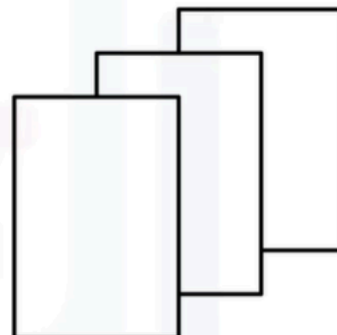
- Relational databases (SQL)
- Spreadsheets



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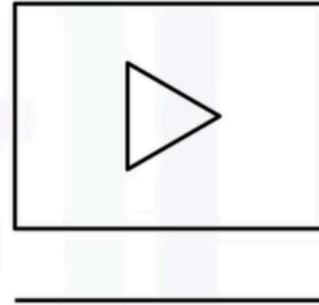
Sources: Semi-structured

- XML
- JSON



Explosion in unstructured data

- Video production
- Social media
- Internet speeds



- Some sources of structured Big Data are relational databases and spreadsheets. With this type of structure, we know how data is related to other data, what the data means, and the data is easy to query using a programming language like SQL.
- Some sources of semi-structured Big Data are XML and JSON files. These sources use tags or other markers to enforce hierarchies of records and fields within data.
- Currently, there is an explosion of data coming from Internet activity, and in particular, video production and consumption as well as social media activities. These numbers will just keep growing as Internet speeds increase, and as more and more people all over the world have access to the Internet.

Advances in cloud computing

- Advances in cloud computing have contributed to the increasing potential of Big Data
- Cloud computing allows users to access highly scalable computing and storage resources through the Internet
- Organizations can expand server capacity to the large scale required to process Big Data



- The emergence of cloud computing has contributed significantly to the launch of the Big Data era: Cloud computing allows users to access highly scalable computing and storage resources through the Internet. By using cloud computing, companies can use server capacity as needed, and expand it rapidly to the large scale required to process big data sets and run complicated mathematical models. Cloud computing lowers the price to analyze big data as the resources are shared across many users, who pay only for the capacity they actually use.